Annu. Rev. Mater. Sci. 1991. 21:23–44
Copyright © 1991 by Annual Reviews Inc. All rights reserved

ANNUAL REVIEWS Further
Quick links to online content

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

CONTAINERLESS UNDERCOOLING AND SOLIDIFICATION OF PURE METALS

D. M. Herlach

Institut für Raumsimulation, Deutsche Forschungsanstalt für Luft- und Raumfahrt, D-5000 Köln 90, Germany

KEY WORDS: thermodynamics of undercooled melts, thermophysical properties of undercooled melts, nucleation, dendritic growth, phase formation

INTRODUCTION

Containerless processing is an effective tool for undercooling metallic melts far below their equilibrium melting temperatures. By using these methods, the generally dominant heterogeneous nucleation on container walls is completely eliminated. Furthermore, if the experiments are performed under clean environmental conditions, heterogeneous nucleation on free surfaces is also greatly reduced. In this way, levels of undercooling can be reached for bulk liquids that are otherwise achieved only by the droplet emulsion technique.

Containerless undercooling and solidification can be realized either by levitation or by drop tubes. In the former case, the gravitational force acting on the sample is compensated for by levitation forces of the electromagnetic fields. In the latter case the containerless state is achieved by the free fall of small droplets into drop tubes.

An undercooled melt is a non-equilibrium state, which is of scientific interest. For example, the degree of undercooling achieved prior to solidification gives information about the efficiency of nucleants. In particular, nucleation is the decisive process selecting the phase, stable or meta-

0084-6600/91/0801-0023$02.00

stable, to be crystallized. Once nucleation has occurred, subsequent growth completes crystallization. In the case of highly undercooled melts, one has to deal with rapid dendrite growth, in which deviations from the local equilibrium at the interface may be expected. Also, the non-equilibrium state of an undercooled melt is of special interest with respect to its thermophysical properties. The study of such phenomena in undercooled melts is less advanced, but considerable recent progress has been made through the application of containerless processing techniques. This is the subject of our review.

TECHNIQUES FOR CONTAINERLESS UNDERCOOLING AND SOLIDIFICATION

Drop Tubes

An established experimental method for containerless processing is the use of drop tubes of various lengths. As an example, at the NASA Marshal Space Flight Center a 32 m drop tube (1) is used. Individual droplets are undercooled during free fall either in a vacuum environment of pressure 10^{-5} mbar or in an Ar or He atmosphere. The sample, supported by a thin wire, is melted by electron beam bombardment. After melting the droplet is released, falls into the drop tube and begins to undercool. The undercooling of the drop will continue until nucleation begins, which will result in rapid crystal growth with a subsequent rapid rise in the temperature of the droplet caused by the release of the latent heat of fusion. The sudden increase of temperature during recalescence is measured by photodetectors arranged at various places along the drop tube. If the heat transfer during fall of the droplet can be calculated, measuring the falling time until recalescence occurs allows one to evaluate the undercooling prior to nucleation.

A similar drop tube was recently built in Grenoble with a total length of 48 m, which makes use of Ultra-High-Vacuum (UHV) conditions (pressure $4 \cdot 10^{-10}$ mbar) (2). An UHV environment greatly lessens the potential heterogeneous nucleation site as surface contamination of the droplet is reduced.

Drop tubes of smaller size are widely used in many laboratories, in particular, for statistical analysis of the solidification behavior of an ensemble of droplets in the range of diameter of 0.1 to 1 mm (3, 4, 5, 6). Usually the sample material is melted in a crucible and, subsequently, dispersed as droplets into the drop tube by forcing the melt by Ar or He gas pressure through a thin nozzle at the bottom of the container. These drop tubes, however, do not allow a recording of thermal history during solidification of the individual droplets.

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

Containerless processing conditions also prevail in melt atomization experiments, in which the melt is atomized by high pressure gases or by electrohydrodynamic fields. Melt atomization decreases the droplet size to diameters of 1 μm or smaller. As the droplet size reaches such dimensions, an effective isolation of heterogeneous nucleants within the volume is expected. Assuming that volume heterogeneities of density P are randomly distributed, the fraction of droplets free of heterogeneous nucleants, X should rise rapidly with decreasing droplet diameter d according to the Poisson distribution function $X = \exp(-P\pi d^3/6)$. In particular, atomizing melts electrohydrodynamically under high vacuum, or even under UHV conditions, is an effective means to eliminate heterogeneous nucleants, since this technique is containerless, works under vacuum conditions, and produces highly refined powders. Actually, experiments using this technique give indirect evidence that the limit of maximum undercooling, as given by the onset of homogeneous nucleation, may be reached for metallic liquids (7, 8).

Electromagnetic Levitation

The application of drop tubes for containerless processing is limited to small droplets because of the short experimental time during free fall. This makes it extremely difficult to perform direct diagnostics during undercooling and solidification. Levitation of bulk samples offers the possibility of undercooling massive samples, which remain accessible not only for direct observation of undercooling and solidification, but also for external stimulation. For metallic systems, the most suitable means for freely suspending spheres in diameter up to 1 cm is the electromagnetic levitation technique. Figure 1 shows a schematic view of an electromagnetic levitation chamber, which is used for undercooling experiments with respect to nucleation, rapid crystal growth, and formation of metastable phases and microstructures (9).

The sample is placed within a conical levitation coil typically consisting of five to seven water-cooled copper windings with one or two counterwindings at its top. The coil is placed within an UHV-chamber and is connected to a high frequency generator of variable power and frequency via an electrical feedthrough. The UHV recipient can be evacuated to pressures of 10^{-9} mbar and backfilled with purified gases such as Ar, He, or He-H mixtures. An inhomogeneous DC magnetic field can be superposed, which serves as a protection against sample instabilities such as rotation and oscillations (10). The temperature of the sample is measured from the top, using a two color pyrometer with an absolute accuracy of ± 5 K and with a sampling rate of 100 Hz. Additionally, a high speed photosensing device arranged in a plane perpendicular to the symmetry

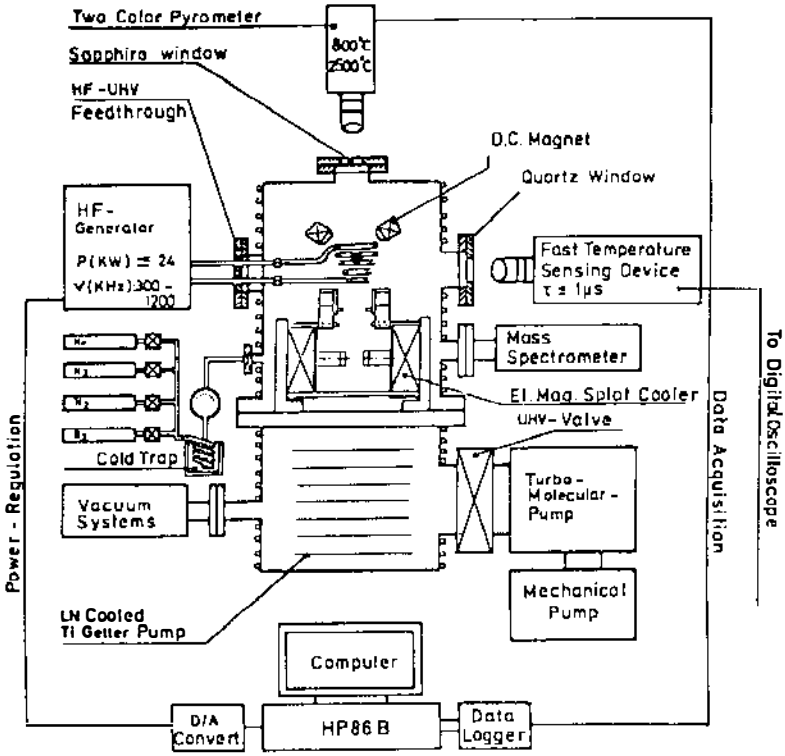

Figure 1 Electromagnetic levitation apparatus for containerless undercooling and solidification of metals. This equipment allows direct measurements of the undercooling by contactless temperature measurements. Time-resolved temperature measurements of the rapid rise of temperature during recalescence give information on the crystallization process (9, 11).

axis of the coil can record relative temperature changes with a sampling rate of 1 MHz (11).

The eddy currents induced in the metallic sample lead to a repulsion force against the primary field and simultaneously induce a heating effect. Assuming symmetric coil loops, a sample diameter much less than the diameter of the coil, and replacing the eddy currents distributed over the whole surface of the sample by a single loop circuit, the levitation force F_L and the power absorption P are given by (12):

$$F_L = \text{const}\, G(x) \cdot B \cdot \text{grad}\, B, \qquad 1.$$

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

$$P = \text{const}\, \kappa^{-1} \cdot H(x) \cdot B^2 \qquad 2.$$

where B is the magnetic field strength; $G(x)$ and $H(x)$ are functions depending, in a different way, on $x = 0.5d/\delta$. δ denotes the skin depth; $\delta = 1/\sqrt{\pi f \mu \kappa}$ with f, the frequency of the alternating electromagnetic field, μ the magnetic permeability, and κ the electrical conductivity of the sample. More recently, the theory of electromagnetic levitation has been expanded to allow analytical calculations of the field distribution of asymmetric realistic coil systems in axial and radial directions (13).

Temperature control of levitated samples requires a separate action of P and F_L as far as possible. As a consequence of Equations 1 and 2, the essential difference between P and F_L is that: (*a*) the functions $G(x)$ and $H(x)$ have different characteristics with respect to the frequency of the field; and (*b*) F_L depends on the product B grad B, while P is proportional to B^2. Hence, temperature control is possible within a limited range by a variation of the sample along the symmetry axis of a conical-shaped coil. In lower regions of the coil, the windings are more narrow, so the magnetic field and power absorption is larger than in the upper region of the coil of lower field strength, where the field gradient is increased. Using coils of proper geometry, controlled temperature variation is possible over some hundred degrees.

The achievable temperature range is limited by the balance of heat produced in the sample and heat transferred to the environment. Due to the strong levitation fields needed for the compensation of the gravitational force under terrestrial conditions, an equivalent high power absorption within the levitated drops will be present, thus leading to a strong heating effect. Under vacuum conditions, heat transfer takes place by radiation only. This fact limits the applicability of the electromagnetic levitation technique under vacuum to undercooling experiments on very high melting point metals such as W or Nb. Additional convective cooling by He gas, for example, makes the technique suitable for undercooling of transition metals such as Fe and Ni (14).

The application of cooling gases may cause problems with impurities on the sample surface by metal oxides for example. Note that a purity of 1 ppb of the cleanest gas corresponds to a partial pressure of impurities on the order of 10^{-6} mbar. This is, however, not an ultra-high purity environment with respect to contamination of the sample surface during the experimental time duration of a few minutes needed for an undercooling experiment.

These limitations, as well as the restrictions concerning temperature range and control, are overcome by using the special environmental conditions of microgravity. In space, the forces required for the compensation

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

of disturbing accelerations are at least three orders of magnitude smaller than under terrestrial conditions. A means, called TEMPUS, for containerless positioning and inductive heating under microgravity has been developed for this purpose (15).

In contrast to the electromagnetic levitation chamber shown in Figure 1, TEMPUS possesses two different coils connected to two independently working solid-state generators. The first is a coil system producing a quadrupole field with large field gradients and small magnetic fields, i.e. with strong positioning forces, but small power absorption. The second is a coil system producing a dipole field with small field gradients (small positioning forces), but high magnetic field strength (effective heating). In this way, positioning and heating can be controlled independently. Moreover, the small power input needed for efficient positioning allows the sample to be cooled to temperatures down to about 600 K by radiation only. This offers the advantage that even low melting metals can be processed under UHV conditions with the extra benefit of surface purification resulting from evaporation of surface layers under vacuum and almost ideal spherically shaped samples up to 1 cm in diameter.

MAXIMUM UNDERCOOLING

Experimental Results

Table 1 summarizes the maximum absolute undercoolings $\Delta T = T_m - T_s$ (T_m melting temperature, T_s solidification temperature) and the maximum relative undercoolings $\Delta T/T_m$ for pure metals processed in the containerless manner by either drop tubes (DT) (16, 17) or electromagnetic levitation (EM) (14, 18, 19, 20).

Table 1 reveals that relative undercoolings are reported in the range of 0.15–0.30 in $\Delta T/T_m$. The largest undercooling has been found in electromagnetically levitated Fe (19). However, this result has not been reproduced thus far by other experimenters.

Such undercoolings have been achieved earlier on small droplets a few μm in size by the dispersion technique (21, 22). Thus, maximum undercoolings of $\Delta T/T_m$ are evidently not limited to small droplets, but also can be achieved in bulk melts. This gives evidence that surface heterogeneous nucleation and not volume nucleation is the decisive process limiting the undercooling of pure metals. The ultimate limit of undercoolability is still in question.

Thermodynamic Constraints

As a melt is undercooled below its melting temperature, a driving force for solidification exists, which is given by the difference of Gibbs free energy between the solid and liquid states, ΔG_v:

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

Table 1 Results of maximum undercooling ΔT as obtained by drop tube (DT) or electromagnetic levitation (EM) experiments

Metal		ΔT (K)	$\Delta T/T_m$	ΔH_f (kJ/mol)	ΔT_{hyp} (K)	d (mm)	$\dot{T}$ (K/s)	ΔT_{hom} (K)	Reference
Ti	DT	325	0.167	15.45	461	1.8	72	549	(16)
Pt	DT	378	0.185	19.6	564	2.6	126	616	(16)
Zr	DT	355	0.167	16.9	505	5.0	101	591	(16)
Rh	DT	457	0.204	21.5		2.3	143	676	(16)
Re	DT	975	0.28	33.2		1			(17)
Hf	DT	438	0.177	24.06	748	4.6	199		(16)
Ru	DT	377	0.15	24.0	—	4.8	99	749	(16)
Ir	DT	477	0.18	26.1		3.3	170	816	(16)
Nb	DT	483	0.176	26.4		4.4	230	664	(16)
Mo	DT	590	0.204	32.0	585	3.8	310	741	(16)
Ta	DT	737	0.226	31.6	—	3.7	567	801	(16)
Fe	EM	324	0.18	13.8	311	6	10	392 (bcc)	(14)
Fe	EM	420	0.23	—	—	6	10	485 (fcc)	(18)
Fe	EM	538	0.30			5	10	--	(19)
Ni	EM	341	0.198	17.47	480	6	10	517	(14)
Ni	EM	452	0.262	---		5	10	518	(19)
Ni	EM	480	0.276	—		6	10	517	(18)
Co	EM	350	0.198	16.19	466	5	10	512	(19)
Cu	EM	266	0.196	13.05	442	6	10	400	(20)

ΔT (K), the relative undercooling $\Delta T/T_m$; the hypercooling $\Delta T_{hyp} = \Delta H_f/C_p^l$ as calculated from data of the heat of fusion ΔH_f and specific heat of the liquid C_p^l (Reference 29); and the undercooling ΔT_{hom} as evaluated within homogeneous nucleation theory (cf text) when drops of diameter d (mm) are cooled with a cooling rate $\dot{T}$ (K/s).

$$\Delta G_v = G_s - G_l = \Delta H - T \cdot \Delta S, \qquad 3.$$

where ΔH and ΔS are the corresponding differences in the enthalpy and entropy, respectively:

$$\Delta H = \Delta H_f - \int_T^{T_m} \Delta C_p \, dT, \qquad 3a.$$

$$\Delta S = \Delta S_f - \int_T^{T_m} \frac{\Delta C_p}{T} \, dT, \qquad 3b.$$

$$\Delta H_f = \Delta S_f / T_m. \qquad 3c.$$

Here, ΔH_f is the enthalpy of fusion; ΔS_f is the entropy of fusion; ΔC_p is the difference of the specific heat between liquid and solid state; and T is the temperature of the undercooled melt. Equation 3 indicates that the

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

specific heat of the undercooled melt is the most decisive parameter in the evaluation of the thermodynamic functions of the melt.

Specific Heat

Up to now, only few data of measurements of the specific heat in the undercooled melt regime have been reported for pure metals (23, 24, 25, 26). Figure 2 shows an example of the specific heat difference as a function of temperature for pure In (23). As the liquid passes into the undercooled regime, no discontinuity appears in $\Delta C_p(T)$. ΔC_p rises linearly with decreasing temperature. The amount of increase in ΔC_p is larger than that suggested by the simple extrapolation of the high temperature data of $T > T_m$. This finding implies a reduction in configurational entropy of the liquid probably because of an increase of topological short range order in the undercooled melt. There have been attempts to describe this behavior within a free volume model by a redistribution of free volume, in which the liquid is described as a solid with a defect concentration much higher than a crystal (27).

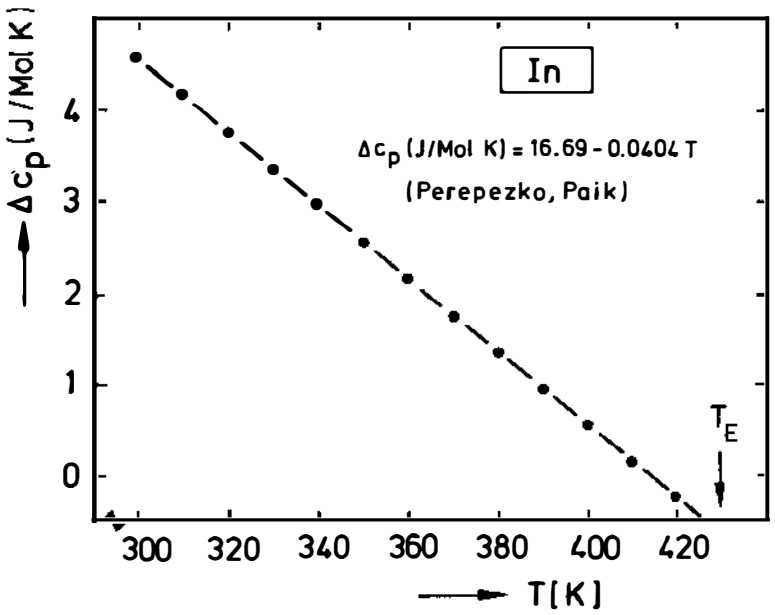

Figure 2 The difference of the specific heat, ΔC_p of In as a function of temperature T in the undercooled melt regime, as measured by Differential Scanning Calorimetry (23). $T_m \equiv T_E$ is the equilibrium melting temperature.

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

Enthalpy

Based on the experimental data on the specific heat difference ΔC_p, an example of which is given in Figure 2 for pure In (23), the difference of the enthalpy between solid and liquid state, ΔH, can be calculated according to Equation 3a. The enthalpy difference represents the heat released upon solidification, which will cause a recalescence of the crystallizing sample. For the case of a highly undercooled melt, a large driving force will exist and, thus, rapid solidification is expected so that adiabatic conditions may be assumed. Under such circumstances, a complete isenthalpic solidification is not possible until the enthalpy of the undercooled liquid at the solidification temperature equals the enthalpy of the solid phase at the melting temperature (cf Figure 3). The temperature that satisfies this isenthalpic crystallization condition, $\Delta H = 0$, marks the onset of the hypercooled liquid regime. The hypercooling limit $\Delta T_{hyp} = T_m - T_{hyp}$ is obtained from

$$\Delta H_f = \int_{T_{hyp}}^{T_m} C_p^l \, dT. \qquad 4.$$

Here, T_{hyp} is the hypercooling temperature at which the enthalpy of the

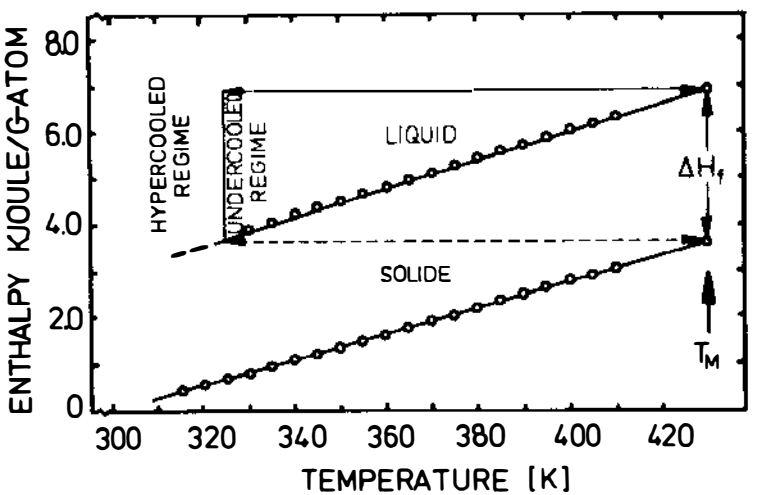

Figure 3 Temperature dependence of the enthalpy for pure In defining the hypercooling onset (23).

undercooled liquid equals the enthalpy of the solid at the melting temperature T_m (Figure 3). The hypercooling limit is relevant to the preparation conditions for metastable phases from the undercooled melt (28), as well as to the kinetics of rapid growth in undercooled melts (see below). Because of the lack of relevant data for the specific heat of the undercooled liquid, there is some uncertainty in the evaluation of the hypercooling limit. The hypercooling values for ΔT_{hyp} are calculated for the different metals listed in Table 1 on the basis of data for the specific heat of the liquid above the melting temperature (29), assuming a constant specific heat below T_m. This is only a rough approach and should give a lower approximation for ΔT_{hyp}, since the specific heat is expected to rise with increasing undercooling (see above). The comparison of the calculated data with the maximum undercoolings as observed by the drop tube or levitation experiments reveals that presumably the hypercooling limit has been reached for only three metals: Mo (DT), Fe (EM), and Ni (EM) (Table 1).

Entropy

As already pointed out by Kauzmann (30), the undercoolability of a melt should be ultimately limited by the condition $\Delta S = 0$. Kauzmann's argument is based on the extrapolation of the entropy of a liquid into the undercooled region. He has shown that the entropy of the undercooled liquid would fall below that of the solid state at the so-called ideal glass temperature (or Kauzmann temperature) T_{og}, which can be obtained from Equation 3b under the condition $\Delta S = 0$. This unphysical condition is avoided if at undercoolings $\Delta T_{og} = T_m - T_{og}$, the undercooled melt undergoes actual freezing into a glassy state.

Recently, the ideal glass temperature T_{og}, according to the Kauzmann paradox, has been calculated for Al on the basis of available C_p data for the crystal, the stable, and the undercooled liquid phase (31). The results of these investigations are shown in Figure 4. The isentropic temperature T_{og} was found to be 225 K at which the entropy of the undercooled liquid is equal to the entropy of the solid. This is $\sim 0.25\ T_m$. Interestingly, a second temperature T_i^s above the melting temperature is postulated, where the condition $\Delta S = 0$ also holds. This has been called the instability temperature of the liquid, where the entropy of the melt again equals that of the solid (for further reading on this phenomenon, see References 31, 32).

Gibbs Free Energy

One of the most important parameters both for nucleation and crystal growth is the Gibbs free energy difference ΔG_v, which can be calculated according to Equation (3). Since relevant data of the specific heat of the

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

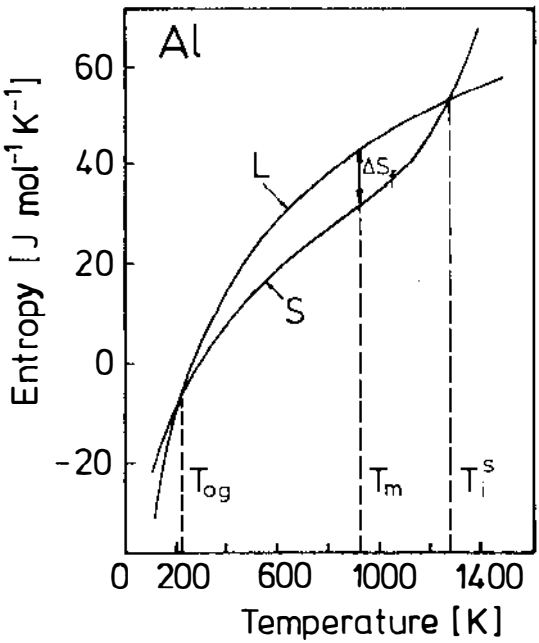

Figure 4 Entropy of liquid and crystalline Al as a function of temperature. T_m is the melting temperature of pure Al (933 K) and T_{og} is the ideal glass temperature where the entropy of the solid equals the entropy of the undercooled liquid. A similar point is found above the melting point, where solid and liquid entropy coincide at T_i^s (31).

undercooled liquid are missing for all but a few metallic systems, different empirical relations have been proposed to describe the temperature dependence of the driving force for solidification. The simplest approximation has been proposed by Turnbull (33). It assumes a zero value of the specific heat difference at the melting temperature, thus according to Equation 3:

$$\Delta G_v = \Delta H_f \cdot \frac{\Delta T}{T_m}. \qquad 5.$$

Equation 5 gives satisfactory agreement for moderate undercooling $\Delta T < 0.20\ T_m$ with data as calculated from the specific heat C_p^l in the undercooled melt region for metals, as far as they are available (23). This is demonstrated in Figure 5, which shows ΔG_v as a function of temperature for the pure metal In, as computed according to different approaches (*solid lines*) for the Gibbs free energy difference and on the basis of measured specific heat values of undercooled In (cf Figure 2). A means of predicting

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

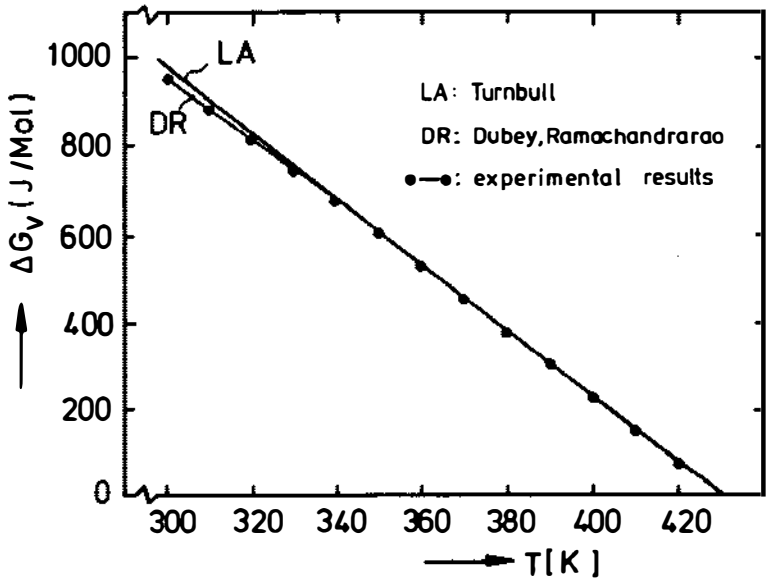

Figure 5 The Gibbs free energy difference ΔG_v as a function of temperature for In: *dots*, calculated on the basis of experimental data for the specific heat and the heat of fusion; *solid line* labeled by LA, prediction of the linear approximation by Turnbull (33); solid line labeled by DR, prediction of the Dubey-Ramachandrarao expression (34).

the Gibbs free energy at even larger undercoolings $\Delta T > 0.2T_m$ [Dubey & Ramachandrarao (34)], was derived on the basis of the free volume model

$$\Delta G_v = \frac{\Delta H_f \cdot \Delta T}{T_m} - \frac{\Delta C_p(T_m) \cdot \Delta T^2}{2T} \cdot \left(1 - \frac{\Delta T}{6T}\right). \qquad 6.$$

Equation 6 allows the determination of ΔG_v over an extended range of undercooling by using only data of the heat of fusion and the specific heat difference at the melting temperature. An accurate prediction of the driving force for nucleation and crystal growth at large undercoolings requires an experimentally measured specific heat of the undercooled liquid.

Nucleation Constraints

Within classical nucleation theory, one may distinguish between homogeneous and heterogeneous nucleation. Homogeneous nucleation is an intrinsic process that depends exclusively on the thermodynamic parameters of the system in question. On the other hand, nucleation can be favored

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

if foreign phases such as container walls or metal oxides participate in the nucleation process. This case corresponds to heterogeneous nucleation, which is extrinsic in nature and can, consequently, be influenced by experimental conditions. Heterogeneous nucleation will generally limit the undercooling of a melt long before homogeneous nucleation is activated.

In order to estimate the maximum undercooling as constrained by the onset of homogeneous nucleation, it is necessary to consider the thermal activation of nuclei. The activation threshold for the formation of a critical homogeneous nucleus of spherical shape is given by (35)

$$\Delta G^* = \frac{16}{3}\pi \cdot \frac{\sigma^3}{\Delta G_v^2}. \qquad 7.$$

Here, σ denotes the interfacial energy between nucleus and melt. This quantity is inaccessible to direct measurements. Spaepen (36) and Thompson (37) developed a structural model that treats the interfacial energy as mainly of entropic origin. This yields

$$\sigma = \alpha \cdot \frac{\Delta S_f}{N_L^{1/3} V_m^{2/3}} \cdot T, \qquad 8.$$

where α is a factor depending on the structure of the nucleus, with numerical values $\alpha = 0.71$ for body centered cubic (bcc), and $\alpha = 0.86$ for face centered cubic (fcc) and hexagonal centered cubic (hcp) structure, respectively. N_L is Avogadro's number and V_m is the molar volume.

Critical and postcritical nuclei arise by addition of single atoms or small groups of atoms to subcritical clusters. This process is assumed to be diffusion controlled (38). Under the assumption of the Einstein-Stokes relation $D = (k_B T)/(6\pi\eta a_o)$: ($k_B$, Boltzmann constant; D, self-diffusion coefficient; η, the temperature-dependent viscosity; and a_o, a typical atomic diameter) Turnbull estimated the homogeneous nucleation rate to (39)

$$I_{ss}^{hom} = \frac{10^{36}}{\eta(T)} \cdot \exp\left[-\frac{\Delta G^*}{k_B T}\right]. \qquad 9.$$

The maximum undercooling ΔT_{hom} as given by the onset of homogeneous nucleation can be estimated on the basis of Equation 9 in combination with Equations 7 and 8. Assuming that a single nucleation event is necessary for the onset of solidification, I_{ss}^{h}
tallization (where V_s is the volume of the sample and t is the observation time). This assumption is fulfilled in pure metals because these materials show a large growth velocity (see below), and once a single nucleation event has occurred, the solidification takes place by subsequent growth only. The observation time can be approximated by $t = \Delta T/\dot{T}$ (with $\dot{T}$ the

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

average cooling rate). Taking into account the diameter of the sample and the cooling rate as given in Table 1, the maximum undercooling for homogeneous nucleation has been calculated (see column ΔT_{hom} Table 1).

A comparison of the calculated values of ΔT_{hom} with the experimentally observed undercoolings in containerlessly processed drops and droplets reveals that the limiting case of homogeneous nucleation has not been observed. Recently, however, a maximum undercooling of 975 K has been reported in case of pure Re processed in the Grenoble drop tube under UHV conditions. Such a high undercooling may be indicative of the onset of homogeneous nucleation (17). This gives support to the assumption that not only containerless processing conditions, but also an ultra pure environment, is a mandatory precondition for the avoidance of heterogeneous nucleation.

SOLIDIFICATION

Measurement of Growth Velocities

As pointed out above, solidification of pure metals largely undercooled by containerless processing takes place after the first nucleation event by subsequent rapid growth. Recently a method has been developed to directly determine the solidification velocity as a function of undercooling by measuring the rapid temperature rise during recalescence on levitated drops externally stimulated at pre-selected undercooling and surface position (11).

Results for measurements of the solidification velocities as a function of undercooling are shown in Figures 6 and 7, for samples of Ni (40) and Fe (41). Considerable solidification velocities up to 70 m/s are observed at the highest undercooling values. In both samples a monotonic rise of the solidification velocity V with increasing undercooling ΔT is found. In the case of Ni the temperature dependence of V changes at a critical undercooling of about $\Delta T^* = 175$ K. Such an apparent break in the growth conditions was observed earlier by Walker (42).

Present experiments concern growth in an undercooled melt, in which the release of the heat of fusion during crystallization will establish a negative temperature gradient in front of the solid-liquid interface. This, in turn, will give rise to an instability of the solidification front leading to dendritic growth.

Theory of Dendritic Growth in Pure Metals

Current theories of dendritic growth in pure metals are based on the solution of the heat transport equation. Considering both the dissipation of the enthalpy of crystallization into the undercooled melt ahead of the sol-

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

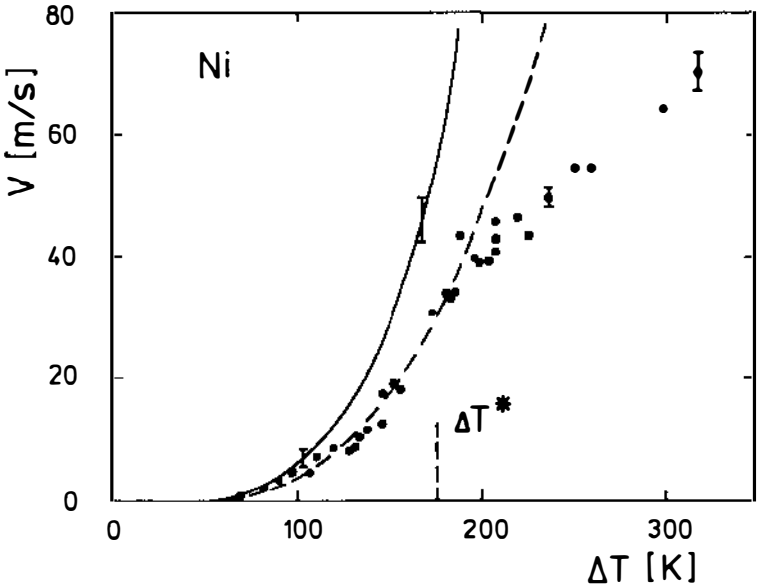

Figure 6 Growth velocity as a function of undercooling for Ni (40). *Dots*, measured values; *solid line*, theoretical prediction according to Reference 43; *dashed line*, theory including non-equilibrium effects at the solid-liquid interface by introducing atomic attachment kinetics.

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

idification front and the Gibbs-Thomson reduction of melting temperature due to the strong curvature at the tip of a dendrite, the total undercooling ΔT can be expressed by (43)

$$\Delta T = \Delta T_t + \Delta T_r = (T_i - T) + (T_m - T_i), \qquad 10.$$

where the thermal undercooling ΔT_t is the difference between the temperature of the dendrite tip, T_i and the melt temperature T far away from the dendrite, and ΔT_r the Gibbs-Thomson undercooling. The individual components are given by

$$\Delta T_t = \frac{\Delta H_f}{C_p^l} P_t \exp(P_t) E_1(P_t) = \Delta T_{hyp} \cdot \mathrm{Iv}(P_t), \qquad 10a.$$

$$\Delta T_r = 2\frac{\Gamma}{R}, \qquad 10b.$$

with $P_t = VR/(2a)$, the thermal Peclet number; V, the dendrite growth velocity; R, the dendrite tip radius; and $a = \lambda/(\rho C_p^l)$, the thermal diffusivity

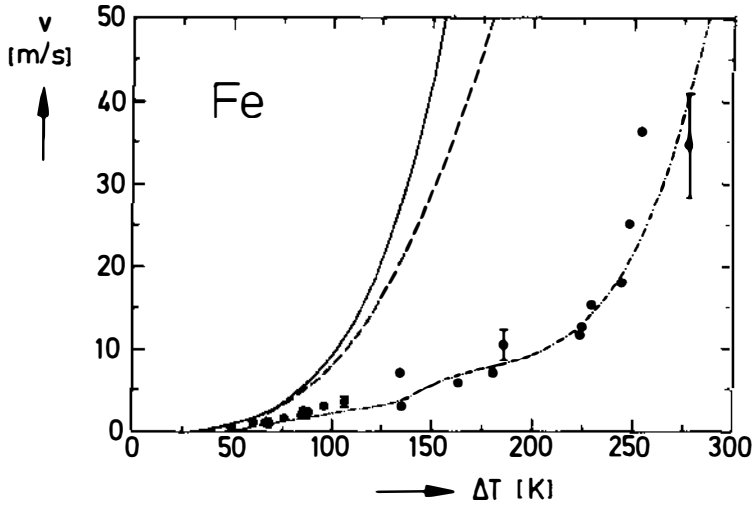

Figure 7 Growth velocity as a function of undercooling for Fe (41). *Dots*, measured values; *solid line*, theoretical prediction according to Reference 43; *dashed line*, theory including an interfacial undercooling; *dash-dotted line*, calculated assuming a strongly partitioning impurity in the melt such as C (0.5 at%).

of the liquid. E_1 denotes the first exponential integral function, Iv the Ivantsov function, and $\Gamma = \sigma/\Delta S_f$, the Gibbs-Thomson coefficient.

Equation 10 gives the correlation between undercooling and the product of VR in terms of the thermal Peclet number. Taking into account the marginal stability criterion (44), the general theory of free dendritic growth yields (45)

$$R = \frac{\Gamma/\sigma^*}{P_t \Delta T_{hyp}(1-\sqrt{(1+1/(\sigma^* P_t^2))})}. \qquad 11.$$

Here, σ^* is a stability constant that can be approximated by $\sigma^* = 1/4\pi^2$. Equation 11 together with Equation 10 allow a unique calculation of the tip velocity V as a function of undercooling ΔT. Using characteristic data for the sample systems measured experimentally (40, 41), the dependence of growth velocity V on undercooling has been computed. In case of Ni and Fe (cf *solid lines* in Figures 6 and 7) the theory leads to an overestimate of the solidification rates.

As pointed out by Coriell & Turnbull (46), reanalysis of the early results of Walker (42) shows that kinetic effects may play a decisive role at high

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

solidification velocities. Following their suggestions, the assumption of local equilibrium at the interface is relaxed by adding a kinetic term $\Delta T_k = V/\mu$ to the right-hand side of Equation 10. Here, $\mu = \Delta H_f V_o/(k_B T_m^2)$ (V_o, velocity of sound) is a coefficient describing the atomic attachment kinetics at the interface (45). The *dashed lines* in Figures 6 and 7 correspond to fits with $\mu = 1.6$ m/(sK). This compares well with values of μ derived from the collision-limited growth model (47) by using known material parameters only (40, 41). In case of Ni, a good agreement with the experiment is achieved at undercoolings less than ΔT^*. In case of Fe, however, considerable deviations remain.

Supposing the Fe sample is contaminated with strongly partitioning elements such as C or B, the constitutional effects both on undercooling and morphological stability may exhibit a strong influence on the calculated $V(\Delta T)$ relation. For example, a concentration of 0.5 at% carbon in iron leads to the *dashed-dotted* line of Figure 7, which can describe the experimental results (41). From this it may be concluded that the measurement of solidification rates is a sensitive detector even for small impurities.

The marked deviations from theoretical predictions observed in the growth behavior of Ni above a critical undercooling ΔT^* are not yet understood. The critical undercooling coincides with the onset of a grain refinement process as evidenced by investigations on alloy systems (48). This has been explained by a fragmentation process of primary formed dendrites, if a critical velocity V^* at ΔT^* is reached, exceeding the atomic diffusive speed.

FUTURE PROSPECTS: THERMODYNAMIC PROPERTIES OF UNDERCOOLED MELTS

The present discussion of nucleation and crystal growth in undercooled metallic melts has demonstrated that a detailed knowledge of thermodynamic parameters such as the specific heat, viscosity, density, and thermal conductivity is a necessary precondition to predict quantitatively the nucleation frequency and crystal growth rate as a function of undercooling. Up to now, little data of such thermodynamic properties measured in the undercooled melt region have been reported. Therefore, efforts have to be directed towards the experimental determination of the temperature-dependence of the thermodynamic properties. In this context, bulk undercooling by electromagnetic levitation both under terrestrial and microgravity conditions offers the possibility to perform such measurements. The following discussion concentrates on the application and necessary modifications of this technique to measure the mass density and thermal

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

expansion, the electrical resistivity, the surface tension and viscosity, and the specific heat and thermal conductivity in the undercooled melt regime.

Mass Density and Thermal Expansion

The electromagnetic levitation technique has been used to measure the mass density of liquid Ni (49) and liquid Cu (50). The change in diameter of a levitated sphere as a function of temperature has been observed by an optical arrangement photographing the profile of the drop. This allowed the measurement of mass density and thermal expansion over an extended temperature range extending far into the undercooled melt region (49). An interesting result has been obtained by extrapolating the experimental data of the undercooled liquid to lower temperatures. This extrapolation suggests that at an undercooling of about 480 K the mass density of the undercooled melt reaches the mass density of the solid. Since metals mostly show a lower mass density in the liquid state, a constraint for maximum undercooling could exist at this undercooling.

Electrical Resistivity

In a spherical shaped sample placed within a rf levitation coil, which is integrated into a free oscillating circuit, the frequency of the coil current will depend on the inductivity of the system consisting of coil and sample material. The inductivity and, consequently, the frequency will change if the electrical resistance of the sample increases with temperature. A method has been developed to apply this principle to measure the temperature change of the electrical resistance of pure Cu and Al with a relative accuracy of better than 0.012% (51).

Specific Heat and Thermal Conductivity

Recently a method was proposed to measure the specific heat of undercooled melts processed in a microgravity environment, using the TEMPUS instrument (52). This instrument consists of two independent coil systems, one for positioning and the other for heating. The heating coil works under steady-state conditions, i.e. there is a constant power input P_o. Steady-state conditions with respect to the temperature of the sample are reached at a temperature T_o, where the absorbed power is equal to the heat transfer to the environment, $P_o = \dot{Q}$. Under UHV conditions, $\dot{Q} = A\varepsilon\sigma_{SB}T^4$ (A, surface area of the sample; ε, emissivity σ_{sb} Stefan-Boltzmann constant). Consider now a modulation of the power input by adding a time-variance power input to the droplet: $P_\omega(t) = P_{\omega 0}\cos^2(\omega t/2)$. This will give rise to temperature oscillations of the form of

$$(\Delta T) = \frac{P_{\omega 0}}{2\omega C_p} \cdot \sqrt{1 + \frac{1}{\omega^2\tau_1^2} + \omega^2\tau_2^2}, \qquad 12.$$

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

where τ_1 is the external (due to heat transfer to the environment) and τ_2 the internal relaxation time (cf Equation 13). Under the condition $\omega\tau_1 > 1 > 10\omega\tau_2$, the heat capacity C_p can be determined from $C_p = P_{\omega o}/2\omega(\Delta T)_\omega$. By measuring $(\Delta T)_\omega$ by contactless pyrometry, selecting and measuring ω by lock-in technique, and $P_{\omega o}$ by calibration, the heat capacity C_p can be determined within an accuracy of 1%.

In a similar way, the thermal conductivity λ is obtained by studying the internal relaxation time, which is given for a droplet of size R by (52)

$$\tau_2 = \frac{3C_p}{4\pi R\gamma\lambda}. \qquad 13.$$

Here γ denotes a geometry factor of one order.

Combining measurements of thermal conductivity and electrical resistance ρ_e allows one to determine the Lorenz number L according to the Wiedemann-Franz law: $\lambda = LT/\rho_e$. Since the Lorenz number crucially depends on the electronic states, such comparative studies give insight into the electronic structure of undercooled metal liquids.

Surface Tension and Viscosity

In the previous section temperature oscillations of a levitated drop have been discussed. Consider now surface oscillations of a freely suspended liquid drop excited by a modulation of the positioning or levitation field. Measuring the frequency of the surface oscillations allows for the determination of the surface tension (19, 53, 54, 55). The frequency υ_l of mode l is correlated with the surface tension σ via (55)

$$\upsilon_l = \sqrt{\frac{l(l-1)(l+2)}{3\pi} \cdot \frac{\sigma}{M}}, \qquad 14.$$

where M is the mass of the sphere. Fourier analysis of the frequency spectrum delivers the oscillation frequencies of mode l. Usually, if the shape of the sample corresponds to an ideal sphere, the l-modes are degenerate. On the other hand, if deviations from the ideal spherical shape are present, the fundamental $l = 2$ mode is split into three sidebands. This has been confirmed by a comparative study of surface oscillations on samples processed under terrestrial and microgravity conditions (56). The strong magnetic fields needed for levitation of liquids in the earth laboratory lead to an asphericity of the sample shape, while microgravity conditions preserve the ideal spherical shape thus allowing for improved accuracy in the determination of the surface tension.

Simultaneously, the investigations of surface oscillations give access to a measurement of the viscosity of the undercooled liquid. The damping

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

behavior of the surface oscillations of the first mode correlates directly to the viscosity η by (57)

$$\eta = R^2/(5\Lambda), \qquad 15.$$

where Λ is the damping factor. The viscosity of undercooled melts will be studied through the damping behavior of oscillating drops processed in TEMPUS during the NASA International Microgravity Laboratory Spacelab mission scheduled for 1993 (58).

SUMMARY AND CONCLUSIONS

Techniques for containerless processing of metals have been described with special emphasis on large drop tubes and electromagnetic levitation that directly study undercooling and solidification of individual droplets and drops. The constraints of maximum undercooling have been outlined on the basis of both thermodynamics and homogeneous nucleation theory. It has been concluded that containerless processing under an ultra-pure environment may give access to the possibility of observing homogeneous nucleation in pure metallic melts.

Measurements of dendrite growth velocities in levitation undercooled metals have been presented, which allow for the verification of current dendrite growth theories. At undercoolings less than critical, there is excellent agreement between the experimental results and the predictions of the theory, in which the solidification velocities can be calculated without any free parameter. These studies show the significance of non-equilibrium conditions at the solid-liquid interface. In the case of pure metals, these conditions lead to an interfacial undercooling. For alloys, additional deviations from the chemical equilibrium at the interface have been observed that are of great importance to solidification into supersaturated solid solutions (59).

The solidification of undercooled melts is governed by nucleation and subsequent growth; nucleation preselects the crystallographic phase, stable or metastable, while growth controls the microstructure evolution. Both processes are of fundamental interest in the production of metastable crystallographic and refined microstructural phases. This is, in particular, true for alloy systems, which show a broad spectrum of possible solidification paths. In such systems, bulk undercooling even offers the potential of seeding metastable phases externally (60).

A quantitative description of the nucleation and growth behavior needs relevant data of thermophysical properties, particularly in the undercooled melt region. Few results of such measurements have been reported up to now. The potential of the electromagnetic levitation technique with

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

emphasis on its use under microgravity conditions has been explored to measure the temperature-dependence of thermodynamic parameters over an extended undercooling range. In particular, these investigations, in combination with undercooling and solidification studies, give a comprehensive description of phase formation and the development of microstructures from the state of an undercooled melt.

ACKNOWLEDGMENTS

The author wants to express his thanks to B. Feuerbacher for stimulating discussions and carefully reading the manuscript and K. Eckler for support in the preparation of the manuscript.

Literature Cited

1. Lacy, L. L., Robinson, M. B., Rathz, T. J. 1981. *J. Cryst. Growth* 51: 47–60
2. Praizey, J. P., Vinet, B., Bost, J. 1989. *Adv. Space Res.* 8: 12–16
3. Drehman, A. J., Turnbull, D. 1981. *Scr. Metall.* 15: 543–48
4. Shong, D. S., Graves, J. A., Ujiie, Y., Perepezko, J. H. 1987. *Mater. Res. Soc. Symp. Proc.* 87: 17–27
5. Gillessen, F., Herlach, D. M. 1988. *Mater. Sci. Eng.* 97: 147–51
6. Cochrane, R. F., Evans, P. V., Greer, A. L. 1988. *Mater. Sci. Eng.* 98: 99–103
7. Bendersky, L. A., Ridder, S. D. 1986. *J. Mater. Res.* 1: 405–14
8. Lin, H. M., Kim, Y. W., Kelly, T. F. 1988. *Acta Metall.* 36: 2537–50
9. Herlach, D. M., Willnecker, R., Gillessen, F. 1985. *Proc. Eur. Symp. Mater. Sci. Microgravity, 5th, SchloB Elmau, 1984*, pp. 399–403. ESA SP-222
10. Essmann, U., Kiessig, H. 1979. *Mater. Res. Bull.* 14: 1139–45
11. Schleip, E., Willnecker, R., Herlach, D. M., Görler, G. P. 1988. *Mater. Sci. Eng.* 98: 39–42
12. Fromm, E., Jehn, H. 1967. *Br. J. Appl. Phys.* 16: 653–63
13. Lohöfer, G. 1989. *SIAM J. Appl. Math.* 49: 567–81
14. Willnecker, R., Herlach, D. M., Feuerbacher, B. 1986. *Appl. Phys. Lett.* 49: 1339–41
15. Piller, J., Knauf, R., Preu, P., Lohöfer, G., Herlach, D. M. 1987. *Proc. Eur. Symp. Mater. Sci. Microgravity, 6th, Bordeaux, 1986*, pp. 437–43. ESA SP-256
16. Hofmeister, W., Robinson, M. B., Bayuzick, R. J. 1987. *Mater. Res. Soc. Symp. Proc.* 87: 149–62
17. Vinet, B., Cortella, L., Favier, J. J., Desre, P. 1990. *Appl. Phys. Lett.* Submitted
18. Gomersall, D. W., Shiraishi, S. Y., Ward, R. G. 1965. *J. Aust. Inst. Met.* 10: 220–24
19. Schade, J., McLean, A., Miller, W. A. 1986. In *Undercooled Alloy Phases*, ed. E. W. Collings, C. C. Koch, pp. 233–48. Warrendale: Metall. Soc.
20. Willnecker, R., Herlach, D. M., Feuerbacher, B. 1988. *Mater. Sci. Eng.* 98: 85–88
21. Cech, R. E., Turnbull, D. 1951. *J. Met.* 191: 242–47
22. Turnbull, D., Cech, R. E. 1950. *J. Appl. Phys.* 21: 804–10
23. Perepezko, J. H., Paik, J. S. 1984. *J. Non-Cryst. Solids* 61–62: 113–18
24. Chen, H. S., Turnbull, D. 1968. *Acta Metall.* 16: 369–73
25. Bosio, L., Cortes, R., Defrain, A. 1973. *J. Chim. Phys. Phys. Chim. Biol.* 70: 357–61
26. Takahashi, Y., Kadokura, H., Yokokawa, H. 1983. *J. Chem. Thermodyn.* 15: 65–71
27. Grest, G. S., Cohen, M. H. 1981. *Adv. Chem. Phys.* 48: 455–71
28. Feuerbacher, B. 1989. *Mater. Sci. Rep.* 4: 1–40
29. Brandes, E. A., ed. 1983. *Smithells Metals Reference Book*, London: Butterworth. 6th ed.
30. Kauzmann, W. 1948. *Chem. Rev.* 43: 219–59
31. Fecht, H. J., Johnson, W. L. 1988. *Nature* 334: 50–51
32. Johnson, W. I., Fecht, H. J. 1988. *J. Less-Common Met.* 145: 63–80
33. Turnbull, D. 1950. *J. Appl. Phys.* 21: 1022–28

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06

34. Dubey, K. S., Ramachandrarao, P. 1984. *Acta Metall.* 32: 91–96
35. Turnbull, D., Fisher, J. C. 1949. *J. Chem. Phys.* 17: 71–73
36. Spaepen, F. 1975. *Acta Metall.* 23: 729–43
37. Thompson, C. V. 1983. *Crystal nucleation in easy glass-forming metallic alloys.* PhD thesis. Harvard Univ. 232 pp.
38. Christian, J. W. 1975. *The Theory of Transformations in Metals and Alloys*, New York: Oxford Univ. Press. 2nd ed.
39. Turnbull, D. 1969. *Contemp. Phys.* 10: 473–88
40. Willnecker, R., Herlach, D. M., Feuerbacher, B. 1989. *Phys. Rev. Lett.* 62: 2707–10
41. Schleip, E. 1990. *Nichtgleichgewichtsevstarrung unterkühlter Metallschmelzen.* PhD thesis. Ruhr-Univ., Bochum. 133 pp.
42. Walker, J. L. 1964. In *Principles of Solidification*, ed. B. Chalmers, p. 114. New York: Wiley
43. Lipton, J., Kurz, W., Trivedi, R. 1987. *Acta Metall.* 35: 957–64
44. Langer, J. S., Müller-Krumbhaar, H. 1978. *Acta Metall.* 26: 1681–88
45. Trivedi, R., Kurz, W. 1986. *Acta Metall.* 34: 1663–70
46. Coriell, S. R., Turnbull, D. 1982. *Acta Metall.* 30: 2135–39
47. Turnbull, D. 1981. *Metall. Trans. A* 12: 693–700
48. Willnecker, R., Herlach, D. M., Feuerbacher, B. 1990. *Appl. Phys. Lett.* 56: 324–26
49. Shiraishi, S. Y., Ward, R. G. 1964. *Can. Metall. Q.* 3: 117–22
50. El-Mehairy, A. E., Ward, R. G. 1963. *Trans. Metall. Soc. AIME* 227: 1226–28
51. Lambeck, M. 1984. *Z. Metallkde.* 75: 806–8
52. Fecht, H. J. 1990. *Rev. Sci. Instrum.* Submitted
53. Fraser, M. E., Lu, W.-K., Hamielec, A. E., Murarka, R. 1971. *Metall. Trans.* 2: 817–23
54. Keene, B. J., Mills, K. C., Kasama, A., McLean, A., Miller, W. A. 1986. *Metall. Trans.* B17: 159–62
55. Eckler, K., Egry, I., Herlach, D. M. 1990. *Mater. Sci. Eng.* Submitted
56. Egry, I., Lohöfer, G., Neuhaus, P., Sauerland, S. 1990. *Int. J. Theromphys.* Submitted
57. Reid, W. H. 1960. *Q. Appl. Math.* 18: 86–88
58. Egry, I., Szcekely, J. 1989. *Propos. Announc. Oppor.* OSSA-4
59. Eckler, E., Cochrane, R. F., Herlach, D. M., Feuerbacher, B. 1990. *Mater. Sci. Eng.* Submitted
60. Schleip, E., Herlach, D. M., Feuerbacher, B. 1990. *Europhys. Lett.* 11: 751–55

Downloaded from www.annualreviews.org. Australia (au) IP: 122.108.64.10 On: Thu, 23 Apr 2026 04:49:06